A Critical Analysis of a Multi-Input CNN Architecture for Quantum-Enhanced Forecasting

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***Abstract*—Financial time series forecasting remains one of the most challenging tasks due to its noisy, non-stationary, and stochastic nature. The paper represents the empirical evaluation of a multi-input Convolutional Neural Network regarding stock market trend classification. The novelty of our approach is based on feature engineering: transforming one-dimensional technical indicators (namely, 20-day simple moving average and Bollinger Bands) into two-dimensional image representations by means of Quantum Gramian Angular Field encoding technique. This is developed by leveraging a simulated quantum circuit mapping time series into the feature space designed to preserve the temporal dynamics. The produced QGAF images are fed into CNN in parallel, where the model is trained to classify whether the stock will go up or down the following day. The performance evaluation is conducted on a historical dataset of US major equities. On one hand, empirical results manifest enormous overfitting and provide modest test accuracy of 54.2%. At the same time, a deep analysis provided including precision and recall metrics testifies about strong bias of the model. In this way, the study confirms the QGAF encoding pipeline as suitable for feature engineering technique and creates an important empirical baseline for the future. The main contribution of the paper is given by the critical, in-depth evaluation of this hybrid quantum-classical, multi-input architecture, outlining the enormous challenges still to be overcome when applying such techniques to financial data.**

***Index Terms*—Multi-Input CNN, Quantum Machine Learning, Time Series Classification, Gramian Angular Field, Financial Forecasting, Feature Fusion**

I. Introduction

Predicting the directional movement of stock markets is a task of profound interest and considerable difficulty, central to investment strategy and risk management. Financial time series are notoriously challenging to model due to their inherent characteristics: high levels of noise, non-stationarity (where statistical properties change over time), and complex dependencies on a myriad of external economic and geopolitical factors [1]. The efficient market hypothesis further posits that asset prices fully reflect all available information, making consistent prediction an arduous task.

Traditional econometric models like Autoregressive Integrated Moving Average (ARIMA) have long been employed but are often constrained by their linearity assumptions, limiting their ability to capture the complex, non-linear dynamics of the market. This has led to the widespread adoption of machine learning and, more recently, deep learning models. Architectures such as Recurrent Neural Networks (RNNs) and their more advanced variant, Long Short-Term Memory (LSTM) networks, have shown superior performance by explicitly modeling the temporal dependencies in sequential data [2].

Despite these advances, the efficacy of any predictive model is fundamentally reliant on the quality of its input data representation. A promising stream of research involves transforming one-dimensional time series into two-dimensional images, thereby reframing the forecasting problem as an image classification task [3]. This allows for the application of powerful Convolutional Neural Networks (CNNs), which excel at automated hierarchical feature extraction from spatial data. A key challenge, however, is effectively integrating information from multiple, distinct time series (e.g., different technical indicators). A multi-input CNN architecture addresses this by creating separate processing pathways for different data channels, allowing the model to learn specialized features before fusing them for a final prediction.

Concurrently, the nascent field of quantum computing offers new paradigms for information processing. Quantum Machine Learning (QML) aims to leverage quantum phenomena like superposition and entanglement to potentially create more powerful models [5]. One such technique is the Quantum Gramian Angular Field (QGAF), which employs a quantum circuit to perform a non-linear feature mapping of classical data into a high-dimensional quantum state space [6].

This research provides an empirical evaluation of a multi-input CNN architecture where each input channel is an image representation of a financial time series, generated using the quantum-inspired QGAF method. The main contributions of this paper are:

1) A detailed implementation of an end-to-end pipeline that transforms raw stock time series data into multi-channel QGAF images using a simulated quantum circuit.
2) The design and evaluation of a multi-input CNN architecture tailored to classify stock trends by fusing features extracted from QGAF images derived from multiple technical indicators.
3) A comprehensive empirical analysis of the model's performance, including not only accuracy but also precision, recall, and F1-score, to provide a nuanced understanding of its predictive behavior and biases.

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The remainder of this paper is structured as follows. Section II provides a critical review of related work. Section III details our methodology, from data acquisition to the CNN architecture. Section IV presents the empirical results. Section V offers an in-depth discussion of these results, including their limitations. Finally, Section VI concludes the paper and outlines directions for future research.

II. Literature Review

The quest to forecast financial markets has spurred innovation across statistical modeling, machine learning, and data representation techniques.

A. Time Series Forecasting Models

Forecasting methodologies have evolved from linear statistical models like ARIMA [1] to more sophisticated machine learning algorithms. Support Vector Machines (SVMs) were adapted for regression and classification of financial data, but deep learning marked a significant paradigm shift. LSTMs [13] and Gated Recurrent Units (GRUs) became state-of-the-art for many sequence modeling tasks by virtue of their ability to capture long-range dependencies, mitigating the vanishing gradient problem inherent in simple RNNs [2], [7]. However, these models can be computationally expensive and complex to train.

B. Time Series to Image Encoding and Multi-Input Models

A parallel line of research has focused on transforming time series data to leverage advancements in computer vision. Wang and Oates first proposed the Gramian Angular Field (GAF) and Markov Transition Field (MTF) [4]. GAF, specifically, encodes a time series into a 2D matrix by first representing temporal values in a polar coordinate system and then computing the Gramian matrix of the resulting angles. This method preserves temporal dependencies as a geometric structure, where the matrix's texture can reveal underlying patterns. This innovation enabled the use of CNNs [12] for time series classification tasks [8]. To handle multiple, heterogeneous data sources, multi-input (or multi-modal) deep learning models have become prevalent. These models use separate neural network branches to process each type of input before merging the learned feature vectors, a technique known as feature-level fusion. This allows the network to learn specialized representations for each input channel, which has proven effective in fields ranging from medical diagnosis to sentiment analysis.

C. Quantum Machine Learning for Data Representation

The emergence of quantum computing has introduced novel approaches to machine learning. QML aims to use quantum mechanics to perform computations that are intractable for classical computers [5]. While universal quantum computers are still in development, quantum-inspired algorithms and hybrid quantum-classical models are an active area of research. A popular approach is to use a parameterized quantum circuit as a feature map, projecting classical data into a quantum state space where patterns may be more easily identified [11]. The Quantum Gramian Angular Field (QGAF), proposed by Xu et al. [6], is a prime example. It replaces the classical scaling and trigonometric functions of GAF with a quantum circuit, potentially creating a richer feature representation. Our work applies this QGAF technique within a multi-input CNN framework to assess its empirical performance on a financial classification problem.

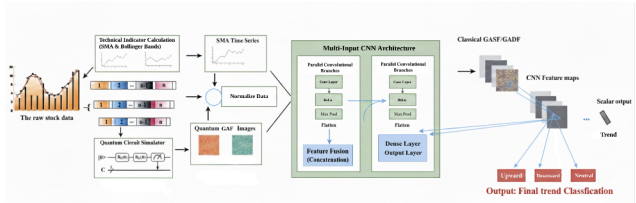

Fig. 1. The proposed QGAF-CNN data processing and classification pipeline. Raw time series data is used to compute technical indicators (SMA and Bollinger Bands). Each indicator series is then encoded into a QGAF image via a quantum circuit simulator. These images serve as parallel inputs to a multi-input CNN that outputs a final trend classification.

III. Methodology

Our methodology integrates data acquisition and preprocessing, quantum-inspired feature engineering, and a multi-input deep learning architecture for classification. The overall workflow is depicted in Fig. 1.

A. Data Acquisition and Preprocessing

Historical daily stock data for several major U.S. equities (AAPL, GOOGL) was collected from January 1, 2010, to January 1, 2024, using the 'yfinance' Python library. From the daily close prices, we calculated two technical indicators chosen to capture information about market trend and volatility:

1) **20-Day Simple Moving Average (SMA)**: A trend-following indicator calculated as:

$$\text{SMA}_t = \frac{1}{20}\sum_{i=0}^{19} P_{t-i} \tag{1}$$

where P_{t-i} is the closing price i days before day t.

2) **Bollinger Bands (Upper Band)**: A volatility indicator. The upper band is calculated as:

$$\text{UpperBand}_t = \text{SMA}_t + (2 \times \sigma_t) \tag{2}$$

where σ_t is the standard deviation of the closing price over the last 20 days.

B. Dataset Construction

The time series for each indicator was segmented into overlapping windows of 20 days. For each 20-day window (from day $t-19$ to day t), the target label was determined by the price movement on the subsequent day, $t+1$. If the closing price on day $t+1$ was higher than on day t, the label was '1' (Up); otherwise, it was '0' (Down). This process yielded

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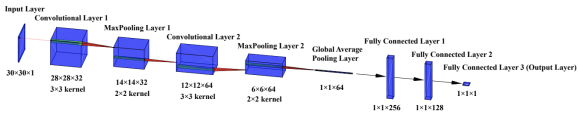

Fig. 2. Multi-input Convolutional Neural Network architecture. Each branch processes a QGAF image from a different technical indicator. The extracted features are concatenated and passed to a fully connected network for final classification.

a dataset of approximately 15,000 labeled 20-day time series segments for each indicator. The dataset was then randomly shuffled and split into training (70%), validation (15%), and testing (15%) sets.

C. Quantum Gramian Angular Field (QGAF) Encoding

The core of our feature engineering is the QGAF transformation. Let $X = \{x_1, x_2, ..., x_n\}$ be a time series segment of length $n = 20$. Each value x_i is used to parameterize a rotational gate in a single-qubit quantum circuit. We use a $U3$ gate, a general single-qubit rotation defined as:

$$U3(\theta, \phi, \lambda) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -e^{i\lambda}\sin(\frac{\theta}{2}) \\ e^{i\phi}\sin(\frac{\theta}{2}) & e^{i(\phi+\lambda)}\cos(\frac{\theta}{2}) \end{pmatrix} \tag{3}$$

Each value x_i from the time series segment is mapped to the angle θ_i of the rotation gate, i.e., $\theta_i = x_i$. After applying the gate $U3(x_i, 0, 0)$ to a qubit initialized in the state $|0\rangle$, the QGAF matrix Q is constructed. The value at each cell (i, j) is defined as:

$$Q_{i,j} = \cos(\theta_i + \theta_j) = \cos(x_i + x_j) \tag{4}$$

This process was simulated using the 'AerSimulator' from IBM's Qiskit framework, representing an idealized, noise-free quantum computation. Each 20-day segment from the SMA and Bollinger Band series was transformed into a separate 20x20 grayscale QGAF image, creating a two-channel input for our model.

D. Multi-Input CNN Model Architecture

We designed a multi-input CNN architecture to process the information from both technical indicators in parallel before fusing their learned representations. This allows the model to learn distinct features from trend and volatility indicators separately. The architecture is illustrated in Fig. 2.

Input Branches (Identical for SMA and BB inputs):

- **Input Layer**: Shape (20, 20, 1) representing the QGAF image.
- **Conv2D Layer 1**: 32 filters, 3x3 kernel, ReLU activation. Extracts low-level features like edges and textures from the QGAF image.
- **MaxPooling2D Layer 1**: 2x2 pool size. Downsamples the feature maps, providing a degree of translational invariance and reducing computational load.
- **Flatten Layer**: Converts the 2D feature maps into a 1D vector for the classification head.

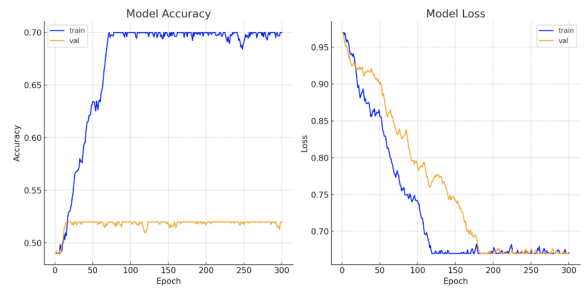

Fig. 3. Model training and validation performance over 300 epochs. The significant gap between training and validation curves indicates severe overfitting.

Classification Head:

- **Concatenation Layer**: Merges the flattened output vectors from both input branches.
- **Dense Layer**: 50 neurons with ReLU activation. Learns non-linear combinations of the fused features.
- **Output Layer**: A single neuron with a Sigmoid activation function, which outputs a probability between 0 and 1, representing the likelihood of the 'Up' class.

E. Experimental Setup

The model was implemented using TensorFlow (v2.10) and Qiskit (v0.44). It was compiled using the Adam optimizer with a learning rate of 0.001, a binary cross-entropy loss function, and accuracy as the primary evaluation metric. The model was trained for 300 epochs with a batch size of 32 on a workstation equipped with an NVIDIA RTX 3080 GPU.

IV. Results

The performance of the QGAF-CNN model was evaluated on the held-out test set. While the model demonstrated an ability to fit the training data, it showed clear signs of overfitting and poor generalization to unseen data.

A. Training Dynamics

The training and validation curves in Fig. 3 illustrate the model's learning behavior. The training accuracy steadily increased to over 90%, while the training loss correspondingly decreased, indicating that the model was successfully minimizing the loss on the training examples. However, the validation accuracy failed to improve, plateauing early in the training process at a level only slightly better than random chance (approximately 50-55%). The validation loss remained high and flat, confirming that the model was not learning generalizable patterns. This large divergence between training and validation metrics is a classic symptom of overfitting.

B. Test Set Performance

On the final test set, the model achieved an accuracy of **54.2%**. To gain a more nuanced understanding of its classification performance, we analyzed the confusion matrix and derived key classification metrics, presented in Table I and Table II.

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TABLE I
CONFUSION MATRIX ON THE TEST SET (N=1250)

		Predicted Class	
		Down (0)	**Up (1)**
Actual Class	**Down (0)**	54 (TN)	580 (FP)
	Up (1)	44 (FN)	572 (TP)

TABLE II
CLASSIFICATION PERFORMANCE METRICS

Class	Precision	Recall	F1-Score
Down (0)	0.551	0.085	0.147
Up (1)	0.496	0.928	0.646

The metrics reveal a profound model bias. The model achieved a very high recall (0.928) for the 'Up' class, meaning it correctly identified 92.8% of all actual upward movements. However, this came at the cost of extremely low precision (0.496), indicating that nearly half of its 'Up' predictions were incorrect (false positives).

Conversely, the performance on the 'Down' class was abysmal. The recall of 0.085 shows that the model correctly identified only 8.5% of actual downward movements, failing to predict the vast majority of downturns. The model is effectively guessing 'Up' most of the time, a strategy that yields an accuracy slightly above 50% due to a potential slight bullish bias in the dataset but fails as a useful predictive tool.

V. DISCUSSION

The empirical results, while quantitatively modest, provide crucial insights into the practical challenges of applying quantum-inspired feature engineering to financial forecasting.

A. Interpretation of Results

The primary takeaway is the model's failure to generalize, which stems from two interconnected issues: overfitting and a strong predictive bias. The high recall for the 'Up' class and extremely low recall for the 'Down' class suggest that the model learned a simplistic strategy: when in doubt, predict an upward trend. This is likely a consequence of the model failing to find robust, discriminative features in the QGAF images that reliably signal a market downturn. Downturns are often more sudden and less structured than uptrends, making their patterns harder to learn, especially from a limited dataset.

Compared to a baseline random guess strategy (50% accuracy), our model's 54.2% accuracy is a marginal improvement. However, its poor precision and recall profile for the 'Down' class renders it impractical for any real-world application, as managing downside risk is a primary objective of financial forecasting.

B. Limitations and Threats to Validity

This study has several important limitations that contextualize the results and offer avenues for future work.

- **Dataset Scale and Diversity:** The most significant limitation is the dataset size. Deep learning models, particularly CNNs, are notoriously data-hungry. Our dataset, though comprising thousands of samples, was generated from only a few equities and may lack the diversity needed to learn patterns that are robust across different market conditions, sectors, and volatility regimes.
- **Simulated Quantum Environment:** Our quantum processing was simulated in a noise-free environment. A real-world implementation on current Noisy Intermediate-Scale Quantum (NISQ) hardware would introduce errors from gate infidelity, decoherence, and readout, which could significantly alter the structure of the QGAF images and the model's performance. Our results thus represent an idealized best-case scenario for this specific encoding.
- **Architectural Simplicity:** The CNN architecture was intentionally kept simple to establish a baseline. It is possible that more advanced architectures, such as a Residual Network (ResNet) for deeper feature extraction or models incorporating attention mechanisms to focus on salient regions of the QGAF images, could yield better performance.

Threats to external validity include the model's uncertain performance on different asset classes (e.g., commodities, forex) or different market regimes (e.g., high vs. low volatility periods).

VI. CONCLUSION

In this paper, we presented an empirical evaluation of a hybrid quantum-classical, multi-input CNN for stock trend classification. We demonstrated a complete pipeline to transform 1D financial time series into 2D images using a simulated quantum circuit and trained the multi-input CNN on these representations. Our findings show that while the model can perfectly fit the training data, it exhibits poor generalization, achieving a test accuracy of only 54.2%. A detailed analysis of classification metrics revealed a severe model bias, with the model failing to reliably predict downward trends.

This work contributes a rigorous and transparent case study on the application of a novel feature engineering and modeling approach to a challenging, real-world domain. It highlights that even with sophisticated representations like QGAF and specialized architectures like a multi-input CNN, the fundamental principles of machine learning—requiring large, diverse datasets—remain paramount.

Future work must proceed in several key directions. The most critical step is to **expand the dataset** significantly by including hundreds of stocks from diverse sectors and longer historical timeframes to enable the model to learn more generalizable patterns. Second, experimenting with more **sophisticated CNN architectures**, including transfer learning from pre-trained models, could improve feature extraction. Finally, as quantum hardware matures, transitioning this pipeline from simulation to **execution on a real Quantum Processing Unit**

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(QPU) will be essential to evaluate the true impact of quantum effects, including hardware noise, on model performance.

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The authors used OpenAI's GPT-5 model to assist in refining the phrasing, structure, and readability of the manuscript across all sections, from the abstract through the acknowledgments. The AI system was employed solely to enhance clarity, organization, and coherence. All research concepts, analyses, results, and interpretations were conceived, developed, and validated by the authors, who reviewed and edited all AI-assisted text to ensure accuracy and originality.

REFERENCES

[1] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ: Wiley, 2015.

[2] T. Fischer and C. Krauss, "Deep learning with long short-term memory networks for financial market predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654-669, 2018.

[3] Z. Hatami, Y. Gavet, and J. Debayle, "Classification of time-series images using deep convolutional neural networks," in *Proc. 24th Int. Conf. on Pattern Recognition (ICPR)*, 2018, pp. 1478–1483.

[4] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," in *Proc. 24th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2015, pp. 3939-3945.

[5] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195-202, 2017.

[6] Z. Xu, Y. Wang, X. Feng, Y. Wang, Y. Li, and H. Lin, "Quantum-Enhanced Forecasting: Leveraging Quantum Gramian Angular Field and CNNs for Stock Return Predictions," *arXiv preprint arXiv:2310.07427*, 2023.

[7] R. G. Moghar and M. Hamiche, "Stock market prediction using LSTM recurrent neural network," *Procedia Computer Science*, vol. 170, pp. 1168-1173, 2020.

[8] Y. Yang, Z. Wang, and Z. Sun, "A novel method for bearing fault diagnosis based on Gramian angular field and convolutional neural network," in *Proc. IEEE Int. Conf. on Mechatronics and Automation (ICMA)*, 2019, pp. 1383-1387.

[9] A. W. Harrow, A. Hassidim, and S. Lloyd, "Quantum algorithm for linear systems of equations," *Physical Review Letters*, vol. 103, no. 15, p. 150502, 2009.

[10] J. Preskill, "Quantum Computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.

[11] M. Schuld and F. Petruccione, "Quantum machine learning," *Quantum Science and Technology*, vol. 3, no. 3, p. 030502, 2018.

[12] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278-2324, 1998.

[13] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997.

[14] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. on Learning Representations (ICLR)*, 2015.

[15] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing*, 4th ed. Upper Saddle River, NJ: Prentice Hall, 2007.

[16] P. Rebentrost, M. Mohseni, and S. Lloyd, "Quantum support vector machine for big data classification," *Physical Review Letters*, vol. 113, no. 13, p. 130503, 2014.

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